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Declarations and Definitions

Defining declarations, also known as definitions, besides introducing the name of an object, also establish the creation (where and when) of the object; that is, the allocation of physical memory and its possible initialization. Referencing declarations, or just declarations, simply make their identifiers and types known to the compiler.

Here is an overview. Declaration is also a definition, except if:

- it declares a function without specifying its body

- it has an extern specifier, and has no initializer or body (in case of function)

- it is a typedef declaration

There can be many referencing declarations for the same identifier, especially in a multifile program, but only one defining declaration for that identifier is allowed.

For example:

```
/* Here is a nondefining declaration of function max; */  
/* it merely informs compiler that max is a function */  
int max();
```

```
/* Here is a definition of function max: */  
int max(int x, int y) {  
    return (x >= y) ? x : y;  
}
```

```
/* Definition of variable i: */  
inti;
```

```
/* Following line is an error, i is already defined! */  
inti;
```

Declarations and Declarators

A declaration is a list of names. The names are sometimes referred to as